

A SURVEY ON ARTIFICIAL INTELLIGENCE AND DEEP LEARNING TECHNIQUES FOR DIABETIC RETINOPATHY DETECTION AND CLASSIFICATION

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Abstract. Diabetic Retinopathy (DR) is a leading cause of preventable blindness, and early diagnosis remains challenging due to subtle lesion patterns and the need for expert-grade annotations. Recent advances in Artificial Intelligence—particularly Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs)—have significantly improved automated DR screening accuracy. This survey provides a structured analysis of existing machine learning and deep learning approaches, highlights commonly used datasets, preprocessing strategies, and evaluation metrics, and synthesizes comparative results across major studies. In addition to summarizing current progress, the survey identifies limitations including dataset imbalance, poor model interpretability, and privacy concerns in clinical deployment. Future directions emphasize ViT–XAI integration, multi-modal fundus–OCT frameworks, federated ViT architectures, and privacy-preserving learning to enable reliable, scalable, real-time DR diagnosis.

Keywords: Diabetic Retinopathy, Fundus Imaging, Vision Transformer, Convolutional Neural Networks, Explainable AI, Deep Learning, Classification.

1 Introduction:

Artificial Intelligence (AI) has become integral to modern ophthalmology by enabling automated analysis of retinal fundus images for early detection of Diabetic Retinopathy (DR). While traditional machine learning relied on handcrafted features such as texture and morphology, deep learning approaches—particularly CNNs and Transformers—have demonstrated superior lesion-level understanding and generalization. However, despite these advancements, significant challenges remain: models often struggle with class imbalance, lack interpretability required for clinical adoption, and depend heavily on large annotated datasets. Furthermore, inconsistent evaluation metrics across studies make comparative assessment difficult. This survey addresses these gaps by analytically comparing existing AI methods, presenting a taxonomy of DR detection techniques, synthesizing performance outcomes from recent literature, and outlining limitations and research directions for the next generation of DR screening systems.

2 AI Techniques for Diabetic Retinopathy: A Taxonomy (figure.3.1)

This survey follows a structured taxonomy of AI-based DR detection methods :

1. Traditional Machine Learning
 - SVM, Random Forest, KNN
 - Feature-engineered approaches (texture, wavelet, morphological)
2. Deep Learning Models
 - CNN-based classification
 - Hybrid CNN–RNN architectures
 - Ensemble models
 - Attention-based CNNs
3. Vision Transformers
 - ViT
 - Swin Transformer
 - Hybrid CNN–Transformer architectures
4. Explainable AI (XAI)
 - Grad-CAM
 - LIME
 - SHAP
 - Attention rollout maps
5. Privacy-Preserving Learning
 - Differential Privacy
 - Federated Learning
 - Encryption-based DR screening

3 Background and Related Works:

3.1. Developing a Transparent Diagnosis Model for Diabetic Retinopathy Using Explainable AI

Detecting diabetic retinopathy remains a complex task due to inconsistent symptom presentation, subjective judgments by clinicians, and the often opaque reasoning behind AI-based predictions. Recent efforts incorporate interpretable AI, such as pairing CNNs with explanation tools like LIME, to help clinicians visualize how models make decisions. While this approach has yielded better diagnostic clarity and measurable accuracy gains over traditional approaches, the field still requires larger, more diverse datasets and advances in explainable methods to enable broader, real-time use in clinical environments.

3.2. Diabetic Retinopathy Classification With Deep Learning via Fundus Images: A Short Survey

Deep learning models have been leveraged to automatically grade the severity of diabetic retinopathy using large, expertly annotated image collections. Techniques such as regional image splitting, used in advanced architectures like Inception@4, boost lesion detection sensitivity by focusing on localized zones within retinal scans. Performance evaluations revealed that this method not only attained high accuracy but also demonstrated strong alignment with clinical assessments, indicating practical value for deployment. Ongoing work highlights the importance of rigorous multi-center studies and expanding datasets for more robust and generalizable screening systems.

3.3. Optical imaging for diabetic retinopathy diagnosis and detection using ensemble models

Ensemble Deep Learning (DL) models enhance automated Diabetic Retinopathy (DR) detection by addressing the limitations of manual, error-prone screening. A hybrid framework combining VGG19, ResNet50, and InceptionV3 leverages strengths in fine detail capture, complex feature learning, and multi-scale analysis. Extracted features are integrated and passed through a CNN for binary classification. Using a preprocessed and augmented Kaggle fundus dataset with an 80-20 train-test split, the ensemble trained for 25 epochs with Adam optimization and early stopping. While ResNet50 alone reached 98% accuracy, the ensemble achieved 99%, highlighting improved diagnostic capability. The approach enhances medical image interpretation, reduces diagnostic errors, and shows promise for early DR treatment, with future scope in real-time clinical deployment.

3.4.Enhancing Early Detection of Diabetic Retinopathy Through the Integration of Deep Learning Models and Explainable Artificial Intelligence

The evolution of diabetic retinopathy detection has transitioned from machine learning techniques to deep convolutional networks, with increasing emphasis on model transferability and interpretability. Studies testing a wide range of pre-trained architectures on benchmark datasets like APTOS 2019 and EyePACS reveal considerable variation in both classification and efficiency metrics. Though models like DenseNet201 and MobileNetV2 show promise, their effectiveness is hampered by imbalanced data and inconsistent representation of disease severities, which remain pressing issues for future neural network optimizations.

3.5.Diagnosis of Diabetic Retinopathy Using Deep Neural Networks

Deep Learning (DL) techniques tailored for early Diabetic Retinopathy (DR) detection demonstrate superior performance compared to transfer learning and traditional machine learning baselines. Using the Kaggle Aptos 2019 dataset containing 28,000 images across five severity levels, a customized 5-layer CNN achieved nearly 95% accuracy in multi-class detection and 100% in binary classification. Validation on Messidor2 and IDRiD datasets confirmed strong robustness, with Grad-CAM visualizations highlighting lesion areas to improve clinical trust. The framework shows strong potential for real-world DR screening and severity grading, with future work focusing on balanced datasets, real-time deployment, and further optimization for clinical use.

3.6.DeepDiabetic: An Identification System of Diabetic Eye Diseases Using Deep Neural Networks

DeepDiabetic framework introduces a unified deep learning model for classifying DR, DME, glaucoma, and cataract. Using 1,228 fundus images with preprocessing and augmentation, several models were tested, including EfficientNetB0, VGG16, and ResNet152V2 with GRU variants. EfficientNetB0 achieved the best results with 98.76% accuracy, strong precision/recall, and 0.9977 AUC, outperforming existing methods. The system shows robustness for clinical use, with future work aimed at larger datasets and improved generalization.

3.7.Deep Learning in Automatic Diabetic Retinopathy Detection and Grading Systems: A Comprehensive Survey and Comparison of Methods

Deep learning for automatic Diabetic Retinopathy (DR) detection has advanced from traditional ML to CNN-based methods with transfer learning, hybrid models, and a focus on interpretability. Challenges include class imbalance, dataset variability, and

limited data. In this study, 26 pre-trained CNNs were tested on DRGF (APTOS 2019) and DRR (EyePACS) with preprocessing, augmentation, and Grad-CAM. DenseNet201 achieved 81.76% validation accuracy, while MobileNetV2 was preferred for efficiency, reaching 75% grading and 78% detection accuracy. However, performance on multi-class grading remained limited due to imbalance.

3.8.Enhancing Early-Stage Diabetic Retinopathy Detection Using a Weighted Ensemble of Deep Neural Networks

Weighted ensemble learning for Diabetic Retinopathy (DR) detection combines Inception-v3, VGG16, and a custom CNN to improve early-stage diagnosis. Using the APTOS 2019 dataset of 3,662 fundus images, preprocessing involved Ben Graham's enhancement, Gaussian blurring, cropping, and extensive augmentation to tackle imbalance. Based on validation accuracy, the top three models were integrated through weighted averaging. The ensemble achieved 95.06% accuracy, 87.88% precision, 83.78% recall, 85.69% F1-score, and 98.10% AUC, outperforming individual models. The framework ensures reliable DR detection and shows adaptability for domains like agriculture and plant disease analysis.

3.9.Prediction and analysis of risk factors for diabetic retinopathy based on machine learning and interpretable models

Interpretable machine learning for Diabetic Retinopathy (DR) risk prediction was developed using 1,000 early-warning samples from the National Clinical Medical Sciences Data Center. Preprocessing involved handling missing data, outliers, and feature selection with information gain. Among models compared, CatBoost outperformed SVM, RF, XGBoost, and GBT, reaching 82.5% accuracy and 89% AUC. To enhance trust, SHAP analysis identified renal dysfunction, high HbA1c, and nephropathy markers as major risk factors, with certain blood and liver indicators showing negative correlation. Additional influences included BMI, blood pressure, and fatty liver, highlighting the role of kidney, liver, and glucose control in DR prevention.

3.10.Convolutional Neural Network (CNN) approach for diagnosing and classifying the severity of diabetic retinopathy (DR)

CNN-based Diabetic Retinopathy (DR) classification was developed using 80,000+ Kaggle fundus images across five severity levels. The model applied deep convolution layers with dropout, L2 regularization, leaky ReLU, and weighted class balancing to address overfitting. Preprocessing involved color normalization, 512×512 resizing, and real-time augmentation. Training on an NVIDIA K40c GPU achieved 75% accuracy, 95% specificity, and 30% sensitivity on 5,000 validation images. While effective for

general DR detection, the system showed limited performance in distinguishing mild to severe stages, highlighting challenges in fine-grained feature extraction.

3.11.Application of Artificial Intelligence for Classification, Segmentation, Early Detection, Early Diagnosis, and Grading of Diabetic Retinopathy From Fundus Retinal Images: A Comprehensive Review

AI in Diabetic Retinopathy (DR) management covers classification, segmentation, detection, and grading using fundus images. CNNs dominate the field, supported by ML models like SVM and Random Forest. Standard datasets such as EyePACS, MESSIDOR, and IDRiD are enhanced with CLAHE and Gaussian filtering for preprocessing. Popular models like ResNet, U-Net, and InceptionV3 deliver high accuracy, while LIME and SHAP ensure interpretability. Key challenges remain in dataset imbalance, training costs, and clinical validation, with future work pointing to multimodal integration and custom architectures.

3.12.An Insight on the Timely Diagnosis of Diabetic Retinopathy Using Traditional and AI-Driven Approaches

Systematic review of 116 studies (80 analyzed) compares traditional imaging (fundus, OCT) with AI-based approaches for DR detection. ML models (SVM, Random Forest) evolved into CNN-driven methods, achieving >90% accuracy. Federated Learning addresses privacy while XAI tools enhance clinician trust. Evaluation relies on sensitivity, specificity, AUC-ROC, and complexity. Challenges include dataset bias, interpretability gaps, and infrastructure limits. Future research focuses on multi-modal models, stronger XAI, and clinical validation.

3.13.Advancing diabetic retinopathy diagnosis with fundus imaging

Survey of fundus image-based CAD systems emphasizes challenges in detecting microaneurysms, exudates, and hemorrhages. Current methods often depend on single-feature analysis, limiting accuracy and requiring costly tests. Authors recommend multi-trait-driven CAD using deep learning, ensemble models, and high-dimensional features. Key issues include blood vessel segmentation, noise reduction, and computational costs. The study highlights the need for multi-modal frameworks to improve robustness, reliability.

3.14.A Systematic Review on Fundus Image-Based Diabetic Retinopathy Detection and Grading: Current Status and Future Directions

Systematic review explores ML and DL techniques for DR detection and grading from fundus images. Methods include SVM, KNN, RF, ANN, CNN, Autoencoders, and RBMs, supported by texture, wavelet, histogram, and edge features. Approaches are grouped into lesion-based, binary, and multi-level classification, evaluated via accuracy, recall, precision, specificity, F-measure, and AUC. Key limitations are small datasets, manual annotation, binary grading, and poor image quality. Future work calls for large high-resolution datasets, DME detection.

3.15.A Privacy-Preserving Approach Using Deep Learning Models for Diabetic Retinopathy Diagnosis

Study on privacy-preserving deep learning applies ResNet50 and DenseNet169 to the DDR dataset, showing strong performance with minimal (<2.53%) accuracy loss after Fernet encryption. Differential Privacy was tested but faced challenges in tuning epsilon. To improve security, authors propose blockchain-based frameworks for decentralized data access. Findings confirm encryption can protect medical data without compromising accuracy. Future work includes advanced DL, Federated Learning, and Secure Multi-Party Computation for safe and scalable DR diagnosis. The study highlights the growing importance of balancing medical innovation with patient data confidentiality. It also recommends building standardized privacy frameworks to support broader clinical adoption.

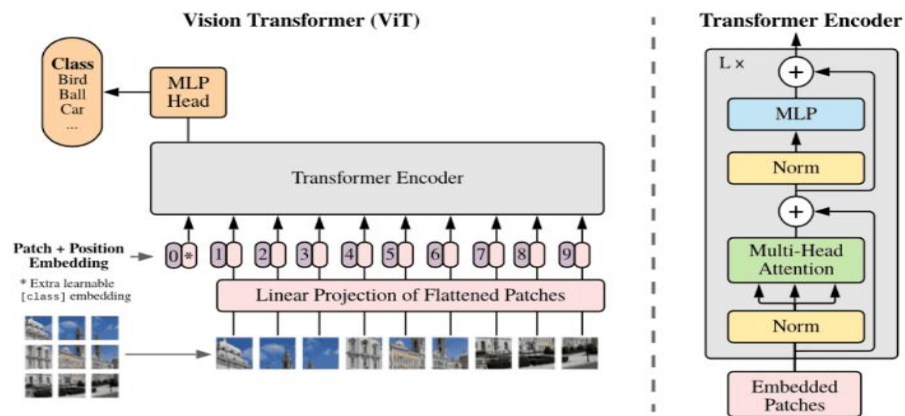


Fig.3.1 Architecture of ViT

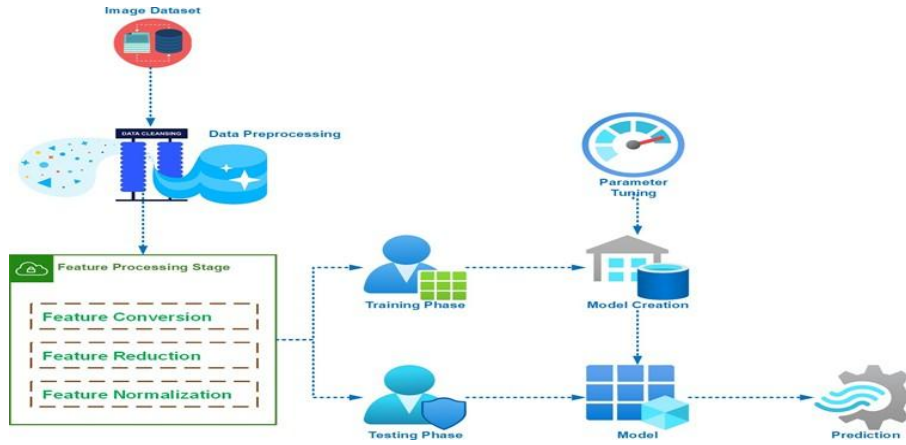


Fig.3.2 Workflow of Proposed system

4 Literature Survey Table:

Ref No	Title	Authors	Methods Used	Advantages	Disadvantages
[1]	Developing a Transparent Diagnosis Model for Diabetic Retinopathy Using Explainable AI	Shahzad et al.	CNN + LIME (XAI)	Improves interpretability, strong diagnostic accuracy	Computationally expensive, small dataset
[2]	Diabetic Retinopathy Classification with Deep Learning via Fundus Images: A Short Survey	Zhu et al.	CNN, ResNet, Inception, DenseNet	Provides concise review of CNN-based approaches	Lacks detailed experiments, limited coverage
[3]	Optical Imaging for Diabetic Retinopathy Diagnosis and Detection Using Ensemble Models	Pavithra et al.	Optical imaging + Ensemble ML	Non-invasive, enhances accuracy through ensembles	Requires costly imaging equipment
[4]	Enhancing Early Detection of DR through DL Models and Explainable AI	Alavee et al.	CNN + Ensemble + Grad-CAM, LIME	High accuracy, interpretable outputs	Complex architecture, longer training time

[5]	Diagnosis of Diabetic Retinopathy Using Deep Neural Networks	Gao et al.	InceptionV3, Inception@4	Clinical reliability, lesion-focused detection	Lower accuracy (88.7%) than modern methods
[6]	DeepDiabetic: An Identification System of Diabetic Eye Diseases Using DNN	Albelaihi et al.	EfficientNetB0, VGG16, ResNet152V2, GRU hybrids	Multi-disease detection, very high accuracy (0.9876)	Dataset small (1228 images), limited diversity
[7]	Deep Learning in Automatic DR Detection and Grading Systems: Survey and Comparison	Abushawish et al.	CNN, ResNet, U-Net, GANs	Comparative analysis of existing DR models	Overfitting risks, interpretability challenges
[8]	Enhancing Early-Stage DR Detection Using a Weighted Ensemble of DNNs	Nazir et al.	Inception-v3, VGG16, Custom CNN (weighted ensemble)	High performance (95% accuracy), strong early detection	Training is complex, high computational demand
[9]	Prediction and Analysis of Risk Factors for DR	Wang et al.	CatBoost ML + SHAP	Identifies and explains DR risk factors	Not image-based, limited clinical deployment
[10]	Convolutional Neural Networks for Diabetic Retinopathy	Pratt et al.	CNN (custom, Kaggle dataset)	First 5-class CNN study, high sensitivity (95%)	Overall accuracy 75%, weak on severe cases
[11]	Application of AI for Classification, Segmentation, Detection, and Grading of DR: A Review	Rajarajeshwari et al.	CNN, ML (SVM, RF), Deep ensemble models	Comprehensive survey of AI in DR	Dataset imbalance, long training times
[12]	An Insight on the Timely Diagnosis of DR Using Traditional and AI-Driven Approaches	Asif et al.	Fundus photography, OCT, CNN, ML	Reviews both manual and AI-driven diagnosis	Manual is subjective, OCT is costly

[13]	Advancing Diabetic Retinopathy Diagnosis with Fundus Imaging: A Comprehensive Survey	Prathibha et al.	Fundus imaging + DL-based methods	Broad survey, highlights advancements	Dependent on image quality, limited scalability
[14]	A Systematic Review on Fundus Image-Based DR Detection and Grading	Ikram et al.	CNN, U-Net, ML classifiers	Covers multiple approaches, identifies gaps	Only a review, no experimental validation
[15]	A Privacy-Preserving Approach Using DL Models for DR Diagnosis	Ng et al.	CNN + Federated Learning + Encryption	Protects patient data privacy	Slight accuracy reduction vs central training

5 Conclusion:

This survey demonstrates that while deep learning—particularly CNNs and ensemble models—has substantially improved automated DR detection, challenges remain in scalability, interpretability, dataset diversity, and privacy compliance. Vision Transformers (ViTs) are emerging as strong alternatives, offering superior global attention modeling and improved lesion-localization capability. However, their success depends on large, well-balanced datasets and robust regularization techniques.

Future work should prioritize:

1. ViT–XAI Fusion: Integrating transformer attention maps with Grad-CAM and SHAP for clinically trusted interpretation.
 2. Multi-Modal Learning: Combining fundus, OCT, and patient metadata for holistic DR assessment.
 3. Federated ViT Architectures: Enabling decentralized, privacy-preserving DR screening without sharing patient data.
 4. High-Resolution Benchmarks: Establishing standardized 4K–8K imaging datasets for consistent evaluation.
 5. Domain-General Models: Building models capable of cross-hospital generalization using self-supervised pretraining.
- Addressing these directions will accelerate the development of robust, transparent, and clinically adoptable DR diagnostic systems.

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